

Bias-Aware Negotiation Agent: Training Framework (CL+RL)

An RL-based negotiation agent to detect and strategically respond to cognitive biases (anchoring, confirmation bias, overconfidence) using multimodal debiasing tools (tone, pace, timing, wording).

The core decision is always binary:

Mandatory Mitigation (if outcome harms fairness)

Ethical Leverage (if outcome benefits the user and is fair)

Stage 1 — Supervised Pretraining (Policy Bootstrapping)

Before RL begins, train the model using labeled synthetic dialogues.

Inputs:

1. Text bias markers (anchoring cues)
2. Prosody signals (pause length, tone confidence)
3. Numerical metrics: Vulnerability score, Sophistication score, Anchoring strength, VAI score, Harm direction (Final Price vs Fair Value)
4. Dialogue turn index

Output: Binary decision + recommended action strategy.

Purpose: Prevent random early RL behavior and ensure the agent understands RULE 0 (harm detection) before exploration.

Stage 2 — Curriculum Reinforcement Learning

Instead of exposing the agent to all complexity at once, difficulty increases in three competency-gated stages.

Stage 2A — RULE 0 Mastery (Clear Harm vs Benefit)

Environment: No severe VAI, No edge cases, Final Price clearly above or below Fair Value

Asymmetry ensures missing harm is treated as more severe than over-caution.

Advance when: $\geq 92\%$ accuracy across 500 held-out cases.

Stage 2B — VAIL + Guardrail Learning

Introduce: Severe VAIL cases, High/Low vulnerability combinations, Guardrail 3 cases (Severe VAIL + Benefit)

Key lesson:

High emotional intensity $\neq$ automatic mitigation.

Harm direction always dominates.

Advance when: $\geq 90\%$ accuracy on Guardrail 3 edge cases while maintaining Stage 2A performance.

Stage 2C — Full Multimodal + Edge Cases

Add: Price range midpoint calculations, Context overrides (urgency, sentiment), Full multimodal action space

Action Space Includes:

Tone modulation, Speech pace, Evidence-based wording, Response timing pauses

Multi-Objective Reward: (dynamic weights)

$$R_{\text{total}} = w_1 * R_{\text{decision}} + w_2 * R_{\text{fairness}} + w_3 * R_{\text{outcome}} + w_4 * R_{\text{trust}}$$

Stage 3 — RL from Human Feedback (RLHF)

After synthetic mastery:

Human evaluators rate: Correct bias classification, Quality of mitigation/leverage, Professionalism, Emotional alignment

A reward model (e.g., Bradley-Terry) converts preference comparisons into a learned reward signal.

Critical safeguard: Human feedback cannot override RULE 0.

Raters influence how the agent responds - not whether mitigation triggers.